R109 | Whole-version A/B comparison

Same frozen data, common base and prototype initialization; NOT parameter/FLOP-matched.

Split	Arm	Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	regret (%)
TRAIN	A	71.481824	11.388574	11.400702	11.345073	-0.203521	0.381317	0.389303
TRAIN	B	71.996091	11.387476	11.400702	11.345073	-0.212836	0.371946	0.379300
VAL	A	71.102469	11.416634	11.427978	11.371692	-0.192841	0.394804	0.404713
VAL	B	71.175309	11.417059	11.427978	11.371692	-0.189295	0.398365	0.408363
TEST	A	71.385185	11.373796	11.385299	11.329932	-0.197930	0.385240	0.394037
TEST	B	71.360494	11.373875	11.385299	11.329932	-0.196958	0.386208	0.393914

A_tied_late | parameters=2,305,033 | selected update=37440

SHA256: 2ba5345464b9b022974cc6bf91303a4728376e3b91b59948921bee8122127b96

B_private_late | parameters=10,220,553 | selected update=37440

SHA256: 93821e6c6dd794f5db08d32cc1defe7c4a2006365e6f864d36fe21f607bd0c84

R109 refines each candidate node graph through two layers on shared original H4 before pooling. A_tied_late ties these weights; B_private_late has 21 banks keyed by global method ID, never candidate column, winner or outcome. Both retain candidate conditioning, original two joint layers, score MLP/maximin and original R77 CE/comparison/risk/two-forward RDropObjective. A has 2,305,033 trainables; B has 10,220,553. The original 1,842,953-parameter base is fresh matched, and all late branches start from identical original-layer prototypes. The initial eval function is identical, but complete state hashes/dimensions are not identical. This is not parameter/FLOP-matched, a routing experiment, or an isolated capacity-only control. Original R88B physical TSP node0 role, methodID+34 and geometry/distance remain; 57,600 updates, logical batch1024, seed2. No new route/path fields, labels, donor, teacher, context, targets, calibration, quotas or auxiliary loss. R89A AdamW, R89B Muon and R84A are saved historical references, not rerun or newly parameter/FLOP-matched controls; R94B is optional. NSS uses saved historical R79 weights, not retrained on new90k. Fixed36 true rescues/losses, false picks and aligned population net costs/regret are separate from selection frequency. Native/FP64 ind and legal pools; TRAIN-fitted SBS; ALL task-equal. TEST is development-exposed; one whole-model full18 VAL checkpoint per arm for full TRAIN/VAL/TEST/LIB; no TEST/LIB reselection. Actual standalone GPU-resident-input timing includes complete method-conditioned node refinement, excludes dataset I/O, CPU transfer and solvers, and is not end-to-end dataset latency. Microcaps/eval batch come from the main-witness locked runner, not this reporter. No routing summary is required. No online W&B equality/availability is asserted.

R109 | TRAIN: ALL and18 tasks

Native/FP64 ind; same per-arm lock; mean cost and actual regret remain distinct.

Problem	A Top1 (%)	B Top1 (%)	A mean_cost	B mean_cost	A regret (%)	B regret (%)	B-A cost
ALL	71.4818	71.9961	11.3885743	11.3874755	0.38930	0.37930	-0.0010987
TSP	46.1877	47.1963	8.0038433	8.0044094	0.41778	0.42566	0.0005661
CVRP	48.1358	50.0160	18.0397910	18.0355095	0.67888	0.65403	-0.0042816
ATSP	76.7086	76.9704	1.6398918	1.6398474	0.36558	0.36340	-0.0000444
OVRP	41.8074	43.5025	7.4433151	7.4415329	0.62918	0.60306	-0.0017822
VRPB	80.8543	81.4086	9.3548260	9.3521157	0.38729	0.35701	-0.0027102
VRPL	38.3667	39.0679	12.3919937	12.3912561	0.39094	0.38467	-0.0007376
VRPTW	80.4494	80.6543	18.8122229	18.8108172	0.30312	0.29388	-0.0014056
OVRPTW	83.9556	84.0716	10.1179001	10.1173305	0.23879	0.23308	-0.0005697
OVRPB	88.4926	88.6654	6.3626343	6.3624292	0.25866	0.25468	-0.0002051
OVRPL	42.2593	43.6086	7.4439461	7.4426220	0.62448	0.60569	-0.0013241
VRPBL	80.0790	80.5901	9.4003419	9.3988684	0.39797	0.38130	-0.0014735
VRPBTW	79.7025	79.8963	19.7304879	19.7281809	0.43102	0.42025	-0.0023070
VRPLTW	80.5481	80.6642	18.7680510	18.7674286	0.30055	0.29700	-0.0006224
OVRPBL	88.6383	88.7877	6.3586496	6.3581032	0.26225	0.25277	-0.0005464
OVRPBTW	83.3654	83.4506	10.6234157	10.6230347	0.32573	0.32106	-0.0003809
OVRPLTW	83.8790	83.9642	10.1103668	10.1100195	0.23981	0.23578	-0.0003473
VRPBLTW	79.7963	79.9247	19.7681654	19.7668273	0.42875	0.42115	-0.0013381
OVRPBLTW	83.4469	83.4901	10.6244939	10.6242272	0.32666	0.32294	-0.0002667

ALL task-equal; raw costs across different problems are descriptive, not a common-unit objective.

SBS fitted on frozen TRAIN; Oracle is each instance legal-pool minimum. No TEST/LIB reselection.

R109 | VAL: ALL and18 tasks

Native/FP64 ind; same per-arm lock; mean cost and actual regret remain distinct.

Problem	A Top1 (%)	B Top1 (%)	A mean_cost	B mean_cost	A regret (%)	B regret (%)	B-A cost
ALL	71.1025	71.1753	11.4166336	11.4170588	0.40471	0.40836	0.0004252
TSP	46.5111	46.3111	8.0247349	8.0263984	0.41842	0.44128	0.0016635
CVRP	46.6667	47.6667	18.1602460	18.1598214	0.70906	0.71545	-0.0004246
ATSP	76.3333	77.0889	1.6388469	1.6387184	0.38600	0.37948	-0.0001285
OVRP	41.3778	41.4222	7.4614054	7.4622164	0.62996	0.63705	0.0008110
VRPB	81.2889	81.2000	9.3555604	9.3595278	0.39605	0.44451	0.0039673
VRPL	37.0667	37.1556	12.4320690	12.4313311	0.40032	0.39384	-0.0007379
VRPTW	80.1111	80.2222	18.8189028	18.8190803	0.29712	0.29654	0.0001775
OVRPTW	83.8889	83.8444	10.1432293	10.1423977	0.25750	0.25049	-0.0008315
OVRPB	88.4444	88.6889	6.3588142	6.3572719	0.29146	0.26650	-0.0015422
OVRPL	42.2667	41.6889	7.4366798	7.4378346	0.64213	0.65648	0.0011548
VRPBL	78.8222	78.9111	9.3697403	9.3705011	0.44791	0.45855	0.0007608
VRPBTW	78.7333	78.6222	19.9447896	19.9497820	0.45240	0.47127	0.0049923
VRPLTW	78.9333	78.8667	18.7639451	18.7640086	0.31600	0.31503	0.0000636
OVRPBL	89.3556	89.2889	6.3691702	6.3698327	0.26766	0.27648	0.0006625
OVRPBTW	83.5778	83.5333	10.6603216	10.6602294	0.32224	0.32079	-0.0000922
OVRPLTW	83.5333	83.5556	10.1570894	10.1562316	0.27190	0.26010	-0.0008577
VRPBLTW	79.7333	79.8667	19.7928917	19.7915299	0.42625	0.41992	-0.0013618
OVRPBLTW	83.2000	83.2222	10.6109685	10.6103450	0.35244	0.34679	-0.0006235

ALL task-equal; raw costs across different problems are descriptive, not a common-unit objective.

SBS fitted on frozen TRAIN; Oracle is each instance legal-pool minimum. No TEST/LIB reselection.

R109 | TEST: ALL and18 tasks

Native/FP64 ind; same per-arm lock; mean cost and actual regret remain distinct.

Problem	A Top1 (%)	B Top1 (%)	A mean_cost	B mean_cost	A regret (%)	B regret (%)	B-A cost
ALL	71.3852	71.3605	11.3737958	11.3738751	0.39404	0.39391	0.0000793
TSP	46.9111	47.1778	7.9710843	7.9716493	0.41350	0.42352	0.0005651
CVRP	48.3111	48.9333	18.0097866	18.0106656	0.66739	0.67090	0.0008790
ATSP	76.6667	77.0667	1.6353799	1.6352525	0.36328	0.35623	-0.0001274
OVRP	40.6889	40.6667	7.4460929	7.4466176	0.62541	0.63184	0.0005247
VRPB	80.5556	80.2889	9.3502460	9.3525122	0.38128	0.40613	0.0022663
VRPL	38.1111	37.0000	12.3488162	12.3491558	0.39561	0.39831	0.0003396
VRPTW	81.1333	81.0889	18.6891373	18.6886688	0.28911	0.28745	-0.0004685
OVRPTW	83.8889	83.9111	10.0928840	10.0928959	0.23647	0.23754	0.0000119
OVRPB	88.0889	88.2000	6.3705248	6.3703336	0.29749	0.29164	-0.0001912
OVRPL	42.1556	42.2222	7.4629911	7.4627593	0.61611	0.61005	-0.0002318
VRPBL	79.4000	79.2889	9.3968819	9.3957354	0.44038	0.42405	-0.0011465
VRPBTW	78.8222	78.8889	19.6653594	19.6622780	0.44701	0.42646	-0.0030814
VRPLTW	79.4222	79.2667	18.8031706	18.8040422	0.32801	0.33024	0.0008715
OVRPBL	89.2667	89.3111	6.3756669	6.3752607	0.26801	0.26215	-0.0004062
OVRPBTW	84.8889	84.8000	10.6596579	10.6608290	0.30181	0.31193	0.0011711
OVRPLTW	82.9778	82.8889	10.0851058	10.0856532	0.25910	0.26490	0.0005474
VRPBLTW	79.9111	79.8222	19.7041274	19.7037225	0.44720	0.44126	-0.0004048
OVRPBLTW	83.7333	83.6667	10.6614124	10.6617208	0.31549	0.31584	0.0003085

ALL task-equal; raw costs across different problems are descriptive, not a common-unit objective.

SBS fitted on frozen TRAIN; Oracle is each instance legal-pool minimum. No TEST/LIB reselection.

R109 | Frozen36 recall and population cost

True rescues/losses and false picks, not selection counts; B versus A.

Split	Winners	A recall (%)	B recall (%)	Rescued	Lost	New false	ALL cost delta	regret delta pp
TRAIN	83279	0.953422	2.829045	1708	146	1565	-0.001098713	-0.010003
VAL	4625	0.691892	1.837838	60	7	110	0.000425190	0.003650
TEST	4598	0.674206	1.761635	57	7	105	0.000079284	-0.000123

Split	New false cost	Removed false cost	Target winner cost	Full18 net cost
TRAIN	0.000204105	-0.000044862	-0.000258244	-0.001098713
VAL	0.000352921	-0.000104218	-0.000178643	0.000425190
TEST	0.000245686	0.000001226	-0.000135054	0.000079284

Target36 SHA256: 406ded5a21d8b73c2a41b87dc2bc3cc14330663ae1aa4f59ca6fbfde83770fe4

Target cost contributions are task-equal; fixed-target changes are not full-population net gain.

R89A/R89B/R84A and historical R79 NSS remain saved references in each arm PDF.